

Chapter 144 — Treatment of Advanced Renal Cell Carcinoma

- **Renal cell carcinoma (RCC) is a family of cancers**, not one disease: histologically, biologically and clinically distinct entities arising in the kidney.
- **Burden**: approximately 81,800 new cancers of the kidney and renal pelvis and approximately 14,890 RCC-related deaths in the United States in 2023; patients with metastatic disease account for the **majority of RCC deaths**.
- **About one-third present with synchronous metastases**, and a further **20–40% of clinically localised tumours later metastasise**.
- **Metastatic RCC is almost always fatal**, with a 10-year survival below 5%.
- Therapy has evolved along three avenues:
 - **Targeted agents**, designed from the genetic and molecular changes underlying each subtype, are now integral to metastatic clear cell RCC (ccRCC); **surgery, irradiation and cytokine therapy remain appropriate in selected patients**.
 - **Immune checkpoint inhibitors**, which modulate T-cell function, are active in ccRCC both front-line and after progression on therapy against the vascular endothelial growth factor (VEGF) axis. Dual checkpoint blockade and combinations of programmed death-1/programmed death ligand-1 (PD-1/PD-L1) inhibitors with agents against VEGF or its receptors (VEGFRs) are now **superior to previous standards**.
 - **Comprehensive genome interrogation** has revealed alterations in carbohydrate and amino-acid metabolism and in chromatin remodelling in ccRCC; their contribution is not yet defined, but they are a hoped-for source of new strategies.
- **Non-clear cell RCC**: ccRCC agents are often used, but with **scant prospective evidence**; defining the oncogenic pathways of papillary, chromophobe and other variants is opening subtype-specific targeted approaches.

PROGNOSTIC FACTORS

- **Five-year survival in metastatic RCC is approximately 12%** (Surveillance, Epidemiology, and End Results [SEER] data); most patients die of their disease.
- **Better outcome**: a long interval between diagnosis and metastasis; fewer metastatic sites. **Worse outcome**: poor performance status; lymph node and/or liver metastases.
- Two models are widely used, especially to stratify clinical trials: the **Memorial Sloan-Kettering Cancer Center (MSKCC) model** and the **International Metastatic Renal Cell Carcinoma Database Consortium (IMDC) model**.
- **MSKCC model — cytokine era**: derived from pretreatment factors in 670 patients in chemotherapy or cytokine trials, most treated with interferon- α (IFN- α). Five factors independently predicted poor outcome on multivariate analysis. Later validated in first-line IFN-based therapy and in external cohorts.
- **IMDC model — tyrosine kinase inhibitor (TKI) era**: derived from patients receiving first-line VEGF-targeted agents at United States and Canadian centres; validated by the same group in an independent cohort of 849 patients and **comparable to the MSKCC model**. It confirmed performance status, hypercalcaemia, anaemia and time from diagnosis to treatment, and **added neutrophilia and thrombocytosis**.
- Validated models guide **management decisions** and the **design and interpretation of trials**.

- **Fig. 144.1** (MSKCC survival by risk group): the curves separate early and stay separated; long-term survivors come almost entirely from the favourable and intermediate groups.

MSKCC VERSUS IMDC RISK MODELS — BOX 144.1, TABLE 144.1, BOX 144.2 AND TABLE 144.2

Box 144.1 Adverse Prognostic Factors in 670 Patients Treated With Chemotherapy or Immunotherapy at the Memorial Sloan-Kettering Cancer Center; Table 144.1 Risk Stratification Based on Adverse Prognostic Factors in 670 Patients (Data from Motzer et al., 1999). Box 144.2 Adverse Prognostic Factors in 849 Patients Treated With First-Line Vascular Endothelial Growth Factor-Targeted Therapy; Table 144.2 Risk Stratification in 849 Patients Treated With VEGF-Targeted Therapy (Heng et al., 2013).

Factor	MSKCC (5 factors)	IMDC (6 factors)
Era / derivation	Cytokine era; chemotherapy or immunotherapy trials	First-line VEGF-targeted therapy
Performance status	Karnofsky <80%	Karnofsky <80%
Haemoglobin	< lower limit of normal	< lower limit of normal
Corrected calcium	>10 mg/dL	> upper limit of normal
Lactate dehydrogenase (LDH)	>1.5 × upper limit of normal	Not included
Time factor	Absence of prior nephrectomy	<1 year from diagnosis to VEGF-targeted therapy
Neutrophils	Not included	> upper limit of normal
Platelets	Not included	> upper limit of normal
Favourable (0 factors) — median overall survival (OS)	20 months	43.2 months
Intermediate (1–2) — median OS	10 months	22.5 months
Poor (≥3) — median OS	4 months (3–5 factors)	7.8 months (3–6 factors)

- **What the table is for:** assigning risk group, which now decides first-line regimen (see Current Therapeutic Strategies).
- **Pattern:** three factors are shared; IMDC swaps LDH and prior nephrectomy for neutrophilia, thrombocytosis and a short diagnosis-to-treatment interval. Median survival in each IMDC group is longer than in its MSKCC counterpart.
- **Trap:** poor risk is **three or more** factors in both models; LDH belongs to MSKCC only; the IMDC calcium threshold is the upper limit of normal, not 10 mg/dL.

Concept — Both models score the same two things, host reserve (performance status, anaemia) and tumour-driven systemic effects (calcium, LDH, neutrophils, platelets) plus pace of disease; IMDC is the cytokine-era model re-derived for targeted therapy.

SURGICAL MANAGEMENT OF METASTATIC RENAL CELL CARCINOMA

Cytoreductive Nephrectomy in Patients With Metastatic Renal Cell Carcinoma

- Studied extensively in the **cytokine era**. The impetus was the perception that bulky tumours suppress key components of immunity, and that removing large primaries might bring clinical benefit.
- **Arguments of early proponents:** (1) rare but documented **spontaneous regression of metastases after nephrectomy**; (2) preclinical evidence that **large primary tumours inhibit T-cell function**; (3) **cytokines rarely induce responses in the primary tumour**.
- **Counterweight:** perioperative morbidity and mortality, and a significant proportion who never proceed to systemic therapy, demanded unequivocal evidence of benefit and a way to select patients.
- **Nephrectomy as the sole intervention does not alter outcome** in metastatic disease.

- **Retrospective combined-modality experience (nephrectomy → cytokine):**
 - In the National Cancer Institute (NCI) series of nephrectomy followed by high-dose interleukin-2 (IL-2), responses were modest and **only about half of patients went on to receive IL-2**; rapid postoperative progression and perioperative morbidity were the commonest barriers.
 - A series including Eastern Cooperative Oncology Group (ECOG) performance status 2, multiple sites, and brain or liver metastases saw **fewer than a quarter reach IL-2**.
 - Series restricted to favourable patients (good performance status, minimal comorbidity, no liver or brain metastases) saw most reach systemic therapy, with **high response rates and good survival**.
 - A population-based SEER analysis associated cytoreductive nephrectomy with better overall and cancer-specific survival.
 - Their main contribution was to establish feasibility and lay the **foundation for randomised trials**.
- **Cytokine-era randomised evidence** — the most compelling support for cytoreductive nephrectomy:
 - **SWOG 8949** (Southwest Oncology Group): IFN- α -2b alone versus after cytoreductive nephrectomy. Eligibility: histological diagnosis of any subtype, ECOG 0 or 1, resectable primary, no prior chemotherapy, irradiation or immunotherapy, adequate organ function. **Response rate did not differ; OS improved.**
 - A smaller **European Organisation for Research and Treatment of Cancer (EORTC)** trial of similar design reproduced the survival advantage, and a **combined analysis** was consistent.
 - **Conclusion:** cytoreductive nephrectomy is supported in **carefully selected patients likely to proceed to cytokine therapy**; its role before systemic therapy in non-clear cell subtypes is unclear.
- **Targeted-therapy era — retrospective support:** the practice continued on cytokine-era data. In a retrospective series of patients receiving VEGF-pathway antagonists, prior cytoreduction was associated with longer survival, and the advantage persisted after adjustment for risk factors. National Cancer Database data showed cytoreduction used more often at academic centres and associated with a survival advantage. Both are subject to **selection bias**.
- **CARMENA** (Cancer du Rein Metastatique Nephrectomie et Antiangiogéniques; French Association of Urology): metastatic ccRCC randomised to **cytoreductive nephrectomy followed by sunitinib versus sunitinib alone**; primary objective noninferiority of sunitinib alone, primary endpoint OS.
 - Planned 576 patients; **poor accrual** (450 at 79 European centres over 9 years) led to **early closure**.
 - **Result:** median OS 13.9 months with nephrectomy–sunitinib versus 18.4 months with sunitinib alone (hazard ratio [HR] 0.89) → **sunitinib alone noninferior** in MSKCC intermediate- and poor-risk patients.
 - **Caveat 1 — population:** almost 45% were MSKCC poor risk, a group not normally offered cytoreduction. About 25% of the nephrectomy arm did not receive the intended treatment (7% no nephrectomy; 18% no sunitinib after nephrectomy), and 17% of the sunitinib arm later underwent nephrectomy. In the per-protocol analysis, **noninferiority was no longer demonstrated**.
 - **Caveat 2 — selection:** investigators could exclude patients with low metastatic burden, so the population most likely to benefit may be **under-represented**.
 - **Caveat 3 — power:** stopped with far fewer events than planned, and the interim analysis had **not met the statistical criteria for early stopping**.
- **SURTIME** (Immediate Surgery or Surgery After Sunitinib Malate in Treating Patients With Metastatic Kidney Cancer): metastatic ccRCC randomised to **immediate cytoreductive nephrectomy versus sunitinib followed by nephrectomy**; hampered by low accrual.
 - **OS favoured deferred nephrectomy** (median 32.4 versus 15.0 months; HR 0.57).
 - **Deferred nephrectomy after sunitinib was safe** — supports sequencing surgery after systemic therapy.

- **Synthesis:** CARMENA and SURTIME reiterate that **poor-prognosis patients are unlikely to benefit**. Their bearing on cytoreduction in selected patients with **low metastatic burden and favourable features remains uncertain**, and their relevance is limited because **sunitinib is no longer standard first-line therapy**.

RANDOMISED TRIALS OF CYTOREDUCTIVE NEPHRECTOMY — INCORPORATING TABLE 144.3

Table 144.3 Summary of Outcome in Randomized Studies of Interferon- α Alone or Interferon- α After Cytoreductive Nephrectomy in Patients With Metastatic Kidney Cancer (Flanigan et al., 2001; Mickisch et al., 2001; Flanigan et al., 2004). CARMENA: Méjean et al., 2018.

Trial	Systemic agent	Comparison	Patients	Response rate	Median OS (months)	Verdict
SWOG 8949	IFN- α -2b	Nephrectomy + IFN vs IFN	241	3.3% vs 3.6% (not significant)	11.1 vs 8.1 (P = 0.05)	Nephrectomy improves OS
EORTC	IFN- α	Nephrectomy + IFN vs IFN	85 randomised; 83 eligible	19% vs 12% (P = 0.38)	17 vs 7 (P = 0.03)	Nephrectomy improves OS
Combined analysis	IFN- α	Nephrectomy + IFN vs IFN	331 (324 in the two arms)	6.9% vs 5.7% (P = 0.60)	13.6 vs 7.8 (P = 0.002)	Consistent with individual trials
CARMENA	Sunitinib	Nephrectomy + sunitinib vs sunitinib	450 (576 planned)	—	13.9 vs 18.4 (HR 0.89)	Sunitinib alone noninferior (intermediate/poor risk)
SURTIME	Sunitinib	Immediate vs deferred nephrectomy	99	—	15.0 vs 32.4 (HR 0.57, favouring deferred)	Deferral safe; OS favours deferral

EORTC: the text reports patients randomised, Table 144.3 reports eligible patients.

- **What the table is for:** the evolution from "nephrectomy then systemic therapy" to "systemic therapy first, nephrectomy selectively".
- **Pattern:** in the cytokine trials cytoreduction improved survival **without changing response rate** — its benefit was not mediated through better cytokine response. With an active systemic agent, upfront surgery no longer helped intermediate- and poor-risk patients.
- **Trap:** CARMENA does not show cytoreduction to be harmful in fit, low-burden, favourable patients — they were largely absent from it.

KEY POINTS: CYTOREDUCTIVE NEPHRECTOMY IN METASTATIC RENAL CELL CARCINOMA

- Two randomised trials showed **improved survival with cytoreductive nephrectomy followed by IFN- α** versus IFN- α alone in carefully selected patients.
- Patients with **poor performance status, comorbidity, rapidly progressive disease or brain metastases** are unlikely to benefit.
- Retrospective data suggested benefit in the VEGFR-targeted era, but a **randomised trial showed no advantage in intermediate- and poor-prognosis patients**; it does not address low-burden, favourable patients.

- The authors' practice is to **offer cytoreductive nephrectomy to well-selected patients with metastatic ccRCC.**

Concept — *Cytoreduction helps only patients fit enough, and with disease slow enough, to reach effective systemic therapy; when that therapy is active, operating first on poor-risk patients mainly delays it — hence selection and deferral.*

Resection of Metastases

- Current systemic agents rarely cure; **resection of limited metastatic disease** is associated with long disease-free intervals and OS in some patients.
- **Never evaluated in a prospective randomised trial.** Favourable outcomes may reflect **selection bias, tumour biology, natural history** or other confounders.
- Evidence is from retrospective series. **Isolated pulmonary metastases** are the lesions most often resectable with curative intent. After complete resection of limited disease, **5-year survival is 35–50%** in many series.
- **Favourable predictors:** complete (curative) resection; **solitary metastasis; age <60 years; disease-free interval >1 year;** pulmonary site; smaller size (<4 cm in one series); **metachronous** presentation.
- **Complete metastasectomy remained feasible and predicted better cancer-specific survival even with multiple lesions.**
- **Bottom line:** without prospective proof of survival benefit, resection of isolated lesions is a **reasonable and widely used practice in well-selected patients.**

KEY POINTS: METASTASECTOMY

- Resection of isolated metastases is **appropriate in selected patients;** complete resection may bring long disease-free intervals.
- Improved outcome is associated with **complete resection, solitary lesion, age <60 years, smaller size, pulmonary site and metachronous disease.**
- **No prospective randomised evidence;** benefit may be selection, biology or natural history rather than resection.

Concept — *Every predictor of success after metastasectomy is also a marker of indolent disease, so the operation selects good biology as much as it treats it.*

Palliative Surgery

- **Palliative cytoreductive nephrectomy** — **indications:** intractable pain, haematuria, constitutional symptoms, and paraneoplastic syndromes (**hypercalcaemia, erythrocytosis, secondary thrombocytosis, hypertension**).
- **Non-surgical alternatives are often effective:** pain and hypercalcaemia are managed medically; haematuria by **angioembolisation**.
- **Resection does not always relieve symptoms** — in one series, only about half of patients with hypercalcaemia had a meaningful fall in calcium. Purely palliative nephrectomy is therefore **infrequent but appropriate in some.**
- **Palliative (non-curative) metastasectomy:** solitary brain metastasis; lesions in **weight-bearing bones or joints;** vertebral metastases with **impending spinal cord or radicular compromise.** Usually combined with radiation and/or systemic therapy.

KEY POINTS: PALLIATIVE SURGERY IN ADVANCED RENAL CELL CARCINOMA

- Cytoreductive nephrectomy may relieve **symptoms from the primary** (pain, haematuria) or **paraneoplastic manifestations**.
- Non-surgical options often suffice, so nephrectomy is **infrequently done with purely palliative intent**.
- Metastasectomy, often with radiation or systemic therapy, is used to relieve symptoms or **prevent life-threatening or disabling sequelae**.

SYSTEMIC THERAPY FOR CLEAR CELL RENAL CELL CARCINOMA

- The United States Food and Drug Administration (FDA) has approved **11 drugs or combinations first-line and 9 in subsequent lines** for advanced RCC (Table 144.4); registration endpoints were objective response rate (ORR), progression-free survival (PFS) or OS.
- **Three eras:** the cytokine era → the single-agent targeted-therapy era → the current immune checkpoint era, often combining a checkpoint inhibitor with a targeted agent.

TABLE 144.4 SUMMARY OF US FDA-APPROVED THERAPIES FOR ADVANCED RENAL CELL CARCINOMA

Agent(s)	Approved	Mechanism	Line	Risk groups	Comparator	Endpoint	Route
First-line therapies							
Interleukin-2	May 1992	Cytokine immunotherapy	First	Favourable	None (phase II)	Objective response rate	IV
Sunitinib	Jan 2006	VEGFR, PDGFR	First	Favourable, intermediate	IFN- α	PFS	Oral
Temsirolimus	May 2007	mTOR	First	Intermediate, poor	IFN- α	OS	IV
Bevacizumab + IFN- α	Jul 2009	Anti-VEGF antibody	First	Favourable, intermediate	IFN- α $\pm$ placebo	OS	IV + SC
Pazopanib	Oct 2009	VEGFR, PDGFR, KIT	First	Favourable, intermediate	Placebo	PFS	Oral
Cabozantinib	Dec 2017	VEGFR, MET, AXL	First	Intermediate, poor	Sunitinib	PFS	Oral
Nivolumab + ipilimumab	Apr 2018	Anti-PD-1 + anti-CTLA-4	First	Intermediate, poor	Sunitinib	OS, ORR, PFS	IV + IV
Pembrolizumab + axitinib	Apr 2019	Anti-PD-1 + VEGFR	First	All	Sunitinib	OS, PFS	IV + oral
Avelumab + axitinib	May 2019	Anti-PD-L1 + VEGFR	First	All (PD-L1-positive tumours)	Sunitinib	OS, PFS	IV + oral
Nivolumab + cabozantinib	Jan 2021	Anti-PD-1 + VEGFR, MET, AXL	First	All	Sunitinib	PFS, OS, ORR	IV + oral
Lenvatinib + pembrolizumab	Aug 2021	VEGFR, PDGFR, KIT, FGFR, RET + anti-PD-1	First	All	Sunitinib	PFS, OS, ORR	Oral + IV
Subsequent-line therapies							
Sorafenib	Dec 2005	VEGFR, PDGFR, KIT	Cytokine failure	Favourable, intermediate	Placebo	OS	Oral
Everolimus	Mar 2009	mTOR	VEGFR failure	All	Placebo	PFS	Oral

Agent(s)	Approved	Mechanism	Line	Risk groups	Comparator	Endpoint	Route
Pazopanib	Oct 2009	VEGFR, PDGFR, KIT	Cytokine failure	Favourable, intermediate	Placebo	PFS	Oral
Axitinib	Jan 2012	VEGFR	Second	—	Sorafenib	PFS	Oral
Nivolumab	Nov 2015	Anti-PD-1	Second	All	Everolimus	OS	IV
Cabozantinib	Apr 2016	VEGFR, MET, AXL	Second	All	Everolimus	PFS	Oral
Lenvatinib + everolimus	May 2016	VEGFR, PDGFR, KIT, FGFR, RET + mTOR	Second	All	Everolimus or lenvatinib	PFS	Oral
Tivozanib	Mar 2021	VEGFR	Third	All	Sorafenib	PFS, OS, ORR	Oral
Belzutifan	Dec 2023	HIF inhibitor	Third	All	Everolimus	PFS, OS	Oral

CTLA-4, cytotoxic T-lymphocyte antigen-4; FGFR, fibroblast growth factor receptor; HIF, hypoxia-inducible factor; IV, intravenous; mTOR, mammalian target of rapamycin; ORR, objective response rate; OS, overall survival; PDGFR, platelet-derived growth factor receptor; PFS, progression-free survival; SC, subcutaneous.

- **What the table is for:** matching each approval to its line, risk group and comparator.
- **Pattern:** every first-line approval since 2018 is a checkpoint combination tested against **sunitinib**. Subsequent-line approvals were tested against **placebo, sorafenib or everolimus**. Only IL-2 was approved on phase II response data.
- **Trap:** the regimens restricted to intermediate/poor risk are **nivolumab + ipilimumab, cabozantinib and temsirolimus**. Temsirolimus is IV, everolimus oral. Axitinib alone is second-line; tivozanib and belzutifan are third-line.

Cytokine Era

- **Summary:** with VEGF and mammalian target of rapamycin (mTOR) inhibitors available, **single-agent IFN is no longer used first-line**. Newer agents rarely produce durable responses, so **high-dose intravenous IL-2 remains reasonable initial therapy in carefully selected patients**.
- **Immunological rationale:**
 - **Spontaneous regression** of metastases after nephrectomy was the earliest evidence of immune control; it is thought to reflect T- or B-cell-mediated immunity, is **rare (<1%) and often transient**, yet shaped immunotherapy development.
 - **Cytotoxic T lymphocytes** within resected tumours, and **tumour-associated antigens** serving as human leukocyte antigen-restricted targets for T-cell killing.
- **Sequence of approaches:** nonspecific immune stimulators (cytokines) ± adoptive cellular therapy → allogeneic immunotherapy, vaccines and modulators of T-cell function. Almost all were directed at ccRCC.

Interferons

- Proteins with diverse functions, including **immunomodulation**. IFN-α was among the first cytokines tested: initially leukocyte-derived, then **recombinant** (early to mid-1980s), which allowed higher doses.
- **Objective response 10–15%** in most trials (higher in early reports); **durable complete responses <2%**.
- **No schedule or route is superior**; adding chemotherapy or other cytokines does not improve on single-agent IFN.
- **Randomised evidence of modest benefit:** IFN-α better than medroxyprogesterone (response and OS); vinblastine + IFN-α better than vinblastine alone (response and OS); a meta-analysis of randomised trials showed a survival advantage for IFN-based therapy.
- Given this, and easier administration than IL-2, IFN was the **usual initial agent until VEGF-pathway antagonists** arrived.

Interleukin-2

- **Response rate 15–20%** across NCI and Cytokine Working Group trials and meta-analyses; early reports exceeded 30%.
- **High-dose IL-2: complete regression in 7–9%, and over 60% of complete responders remain disease-free** on long-term follow-up. Some partial responders became resectable and achieved long-term remission.
- **FDA approval 1992**, based on **durable complete responses**.
- **Regimen:** intravenous bolus **600,000 or 720,000 IU/kg every 8 hours**, as tolerated, to a **maximum of 15 doses**.
- **Toxicity — vascular leak syndrome** → hypotension, third-space fluid retention, respiratory compromise, multiorgan damage. Early treatment-related mortality was **2–5%**; it fell with careful selection, intensive monitoring and early intravenous fluids, vasopressors and antibiotics.
- **Lower-dose regimens** (daily subcutaneous, continuous infusion) appeared comparable in single-arm studies, but two randomised trials showed **lower response rates and fewer durable complete responses** → the authors **recommend only high-dose IL-2**.
- **Combinations do not help:**
 - **IL-2 + IFN:** more toxic, no more effective. In a randomised trial of intermediate-dose continuous-infusion IL-2 versus IFN- α versus the combination, the combination gave higher response and 1-year event-free survival but **no survival difference**, with more toxicity.
 - **IL-2 + chemotherapy** (particularly 5-fluorouracil [5-FU]): high early response rates were **not reproduced**.
 - **IL-2 + tumour-infiltrating lymphocytes or lymphokine-activated killer cells:** no reliable benefit; **largely abandoned**.
- **Predictors of benefit: clear cell histology** (non-clear cell patients too few to judge); performance status, number and site of metastases, prior nephrectomy and time from nephrectomy to systemic therapy. Worst outcome: **more than one metastatic site, metastases within 1 year of diagnosis, liver metastases**.
- **Carbonic anhydrase IX (CAIX, G250):** associated with response retrospectively, but a **prospective trial found it not predictive**.

INTERFERON-A VERSUS HIGH-DOSE INTERLEUKIN-2

	Interferon- α	High-dose interleukin-2
Response rate	10–15%	15–20%
Durable complete response	<2%	7–9%; >60% of these durable
Administration	Easier; schedule and route immaterial	IV bolus every 8 hours, maximum 15 doses; high dose only
Key toxicity	Constitutional	Vascular leak syndrome ; early mortality 2–5%
Current role	Not used first-line	Reasonable in carefully selected patients

Concept — Cytokines are nonspecific immune stimulants with low response rates; high-dose IL-2 alone produces durable complete remission, and that single property is why it survives while IFN does not.

Targeted Therapy Era

- VEGF-pathway antagonists are the **paradigm of rational targeted therapy**: oncogenic pathway → rational drug design → structured clinical evaluation.
- **Von Hippel-Lindau (VHL) disease:** in the early 1990s, NCI investigators used **genetic linkage analysis in VHL families** to identify germline *VHL* mutations or deletions. Carriers develop tumours in multiple organs, including **bilateral, multifocal clear cell kidney cancer**.

- **VHL is a classic tumour suppressor gene:** a somatic "**second hit**" inactivating the normal allele is required for tumour formation.
- **VHL lies on the short arm of chromosome 3;** its protein exists as two alternatively spliced variants.
- **VHL protein function:** forms a complex with **elongins B and C** and **cullin-2 (CUL2)** that tags proteins for **ubiquitin-mediated degradation**; its targets include the **α subunits of the hypoxia-inducible factors (HIFs)**.
- **HIF accumulation upregulates** VEGF, platelet-derived growth factor (PDGF), transforming growth factor- α (TGF- α), glucose transporter-1 (GLUT-1) and erythropoietin, which drive development and progression of ccRCC.
- **Sporadic ccRCC:** somatic *VHL* inactivation, by mutation or **promoter hypermethylation**, is found in the **large majority** of tumours → VHL loss is the **central event** in clear cell oncogenesis and the basis for targeting its pathway.

HIF REGULATION — NORMOXIA, HYPOXIA AND VHL LOSS

Intact VHL, normoxia	Intact VHL, hypoxia	VHL mutated or silenced
Prolyl hydroxylation of HIF- α	Hydroxylation impeded	Hydroxylation irrelevant — VHL cannot bind HIF or elongin/CUL2
VHL-elongin-CUL2 binds → ubiquitination → degradation	No degradation → HIF-α accumulates	HIF-α accumulates even in normoxia
Target genes quiet	Physiological angiogenic response	Constitutive VEGF, PDGF, TGF- α , GLUT-1, erythropoietin

- **Fig. 144.4 — drug targets on the VHL pathway:** bevacizumab (antibody) → VEGF ligand; axitinib, pazopanib and AV-951 (tivozanib) → VEGFR; sunitinib and sorafenib → VEGFR and PDGFR; temsirolimus and everolimus → **mTOR, which promotes HIF translation**.

Bevacizumab

- **Humanised monoclonal antibody against VEGF-A;** the first VEGF-pathway antagonist tested in ccRCC.
- **Randomised phase II after cytokine failure** (3 or 10 mg/kg intravenously every 2 weeks versus placebo): 10 mg/kg prolonged PFS; **no OS gain** (crossover permitted); modest response. Well tolerated: **bleeding, hypertension, fatigue, proteinuria**.
- **Bevacizumab + erlotinib** (an epidermal growth factor receptor [EGFR] tyrosine kinase inhibitor): promising single-arm data, but a randomised phase II trial showed **no superiority over bevacizumab alone**.
- **CALGB 90206** (Cancer and Leukemia Group B) and **AVOREN:** first-line bevacizumab + IFN- α versus IFN- α in metastatic ccRCC.
 - **Higher response and longer PFS** with the combination (CALGB 8.5 versus 5.2 months; AVOREN 10.2 versus 5.4 months).
 - **No OS improvement** in either. There was no crossover, but many patients received off-protocol therapy, which confounds OS.
 - More grade 3 hypertension, fatigue, anorexia and asthenia with the combination.
- **Limitations:** no bevacizumab-alone arm, so IFN's contribution is unknown; the combination's PFS resembles that of single-agent bevacizumab in a first-line phase II study; **never compared with VEGFR TKIs**.
- **Current role:** not widely used as a single agent; being explored with **PD-1/PD-L1 inhibitors** because of its favourable toxicity profile.

Sorafenib

- **Oral kinase inhibitor of VEGFR2, PDGFR- β and raf-1.** First activity shown in a randomised discontinuation phase II trial.
- **TARGET** (phase III, after cytokine therapy, versus placebo): **PFS 5.5 versus 2.8 months**; mature intention-to-treat **OS not significantly different** (17.8 versus 15.3 months).
- **First-line randomised phase II versus IFN- α :** more tumour regression, but **PFS identical**.
- **Toxicity:** hypertension, fatigue, rash, **hand-foot syndrome**, diarrhoea.
- **FDA-approved but now infrequently, if ever, used.**

Sunitinib

- **Oral inhibitor of VEGFR2, PDGFR- β , c-KIT and FMS-like tyrosine kinase-3.** Dual VEGF + PDGF blockade is probably synergistic: VEGF blockade disrupts endothelial proliferation; PDGF blockade disrupts **pericyte** function.
- **Dose:** 50 mg daily for 4 weeks of a 6-week cycle (4 on, 2 off).
- Two single-arm phase II trials, mostly after cytokines, gave high response rates → FDA approval.
- **Landmark phase III versus IFN- α , first-line** (750 patients):
 - **PFS 11 versus 5 months** and **response 31% versus 6%** (independent review).
 - **OS 26.4 versus 21.8 months** — a strong trend ($P = 0.051$).
 - Better quality of life than IFN.
- **Toxicity:** diarrhoea, rash and **hand-foot syndrome**, fatigue and asthenia, hypertension; also **bone marrow suppression and hypothyroidism**.
- **Current role:** superseded by PD-1/PD-L1-based combinations, especially in intermediate and poor risk; used in **favourable-risk patients or those unable to receive checkpoint inhibitors**. It is the comparator arm of every modern first-line trial.

Pazopanib

- **Rationale for selective agents:** sunitinib and sorafenib hit many targets, some irrelevant to ccRCC; dose reductions and discontinuations are common. More selective VEGFR inhibitors aim to cut toxicity without losing efficacy (Table 144.5).
- **Pazopanib:** potent VEGFR inhibitor that **retains PDGFR activity**.
- **Phase III versus placebo** (untreated or one prior cytokine; 2:1 randomisation): PFS and response clearly favoured pazopanib, in both treatment-naïve and cytokine-pretreated subgroups. Mild toxicity; mature **OS not significantly better**.
- **COMPARZ** (phase III, first-line, 1110 patients; pazopanib 800 mg daily versus standard sunitinib): **PFS noninferior** (HR 1.05); **OS comparable** (28.4 versus 29.3 months).
 - **More with sunitinib:** fatigue, **thrombocytopenia**, **hand-foot syndrome**.
 - **More with pazopanib:** raised alanine aminotransferase.
 - Quality of life (fatigue; soreness of mouth, throat, hands and feet) favoured pazopanib in the first 6 months.
- **PISCES** (patient-preference crossover, 10 weeks of each drug with a washout): **most patients preferred pazopanib**, though some preferred sunitinib.
- **Bottom line:** a reasonable alternative wherever sunitinib is suitable; **caution in patients at risk of hepatotoxicity**. In the checkpoint era, single-agent use is largely limited to **good-risk first-line** patients.

Axitinib

- **Highly selective oral TKI of VEGFR1, VEGFR2 and VEGFR3.** High response rates in phase II, mostly after cytokines. Grade 3–4 events: **diarrhoea, fatigue, hypertension**, usually manageable.
- **AXIS** (phase III, second-line, 723 patients after one prior regimen of sunitinib, bevacizumab, temsirolimus or cytokines): axitinib 5 mg twice daily (titratable to 10 mg twice daily) versus sorafenib 400 mg twice daily (no escalation).
 - **PFS 6.7 versus 4.7 months** (HR 0.665), by Response Evaluation Criteria in Solid Tumours (RECIST).
 - Largest benefit **after cytokines** (12.1 versus 6.5 months); **modest after sunitinib** (4.8 versus 3.4 months).
 - **OS not different** (20.1 versus 19.2 months).
- **FDA-approved second-line** on the PFS benefit; now a partner for PD-1/PD-L1 inhibitors.

TABLE 144.5 SUMMARY OF SELECTED STUDIES OF SELECTIVE VASCULAR ENDOTHELIAL GROWTH FACTOR RECEPTOR ANTAGONISTS IN METASTATIC RENAL CELL CARCINOMA

Sternberg et al., 2010, 2013; Motzer et al., 2013; Rini et al., 2011.

Trial	Comparison	Population	Patients	ORR (%)	Median PFS (months)	Median OS (months)
Pazopanib phase III	Pazopanib vs placebo	0–1 prior cytokine	435	30 vs 3	9.2 vs 4.2	22.9 vs 20.5
Tivozanib phase III	Tivozanib vs sorafenib	0–1 prior therapies	517	33 vs 23	11.9 vs 9.1	28.8 vs 29.3
AXIS	Axitinib vs sorafenib	Second-line	723	19 vs 9	6.7 vs 4.7	20.1 vs 19.2
COMPARZ	Pazopanib vs sunitinib	Untreated	1110	31 vs 25	10.5 vs 10.2 (investigator); 8.4 vs 9.5 (independent review)	28.4 vs 29.3

- **Pattern:** the selective agents improved PFS over placebo or sorafenib, but **none improved OS**; COMPARZ established **equivalence, not superiority**, with toxicity as the discriminator.
- **Trap:** COMPARZ is a noninferiority trial — pazopanib and sunitinib are interchangeable on efficacy and differ in side-effect profile.

Cabozantinib

- **Rationale:** most patients progress on VEGF-pathway inhibition. **HIF-driven MET overexpression** may confer resistance to VEGF-axis agents, and **MET and AXL** are overexpressed in ccRCC with worse prognosis.
- **Multikinase inhibitor of VEGFR, MET and AXL.**
- **METEOR** (phase III, 658 patients after VEGF/VEGFR therapy; cabozantinib 60 mg versus everolimus 10 mg daily):
 - **Response 21% versus 5%; PFS 7.4 versus 3.8 months** (HR 0.58).
 - **OS 21.4 versus 17.1 months** (HR 0.70) — an OS benefit.
 - Dose reductions were frequent (fatigue, diarrhoea, **palmar–plantar erythrodysesthesia**), but few discontinued permanently.
- **CABOSUN** (randomised phase II, first-line, IMDC intermediate or poor risk; cabozantinib versus sunitinib): **PFS 8.2 versus 5.6 months** (HR 0.66) and higher response → **FDA approval extended to untreated patients.**

Lenvatinib

- **Rationale:** alternative angiogenic drivers, especially **fibroblast growth factor receptors (FGFR1 and FGFR2)**, mediate resistance to VEGF blockade in preclinical models.
- **Earlier FGFR-active agents disappointed:** nintedanib showed modest activity and no advantage over sunitinib. Dovitinib (VEGFR, PDGFR, FGFR), in a phase III trial versus sorafenib after one VEGF and one mTOR agent, gave **no PFS or OS difference**. This dampened enthusiasm for single agents of this class.
- **Lenvatinib:** inhibitor of VEGFR1–3, FGFR1–4, PDGFR- α , RET and KIT.
- **Three-arm randomised phase II after VEGF-targeted therapy** — lenvatinib 24 mg versus everolimus 10 mg versus lenvatinib 18 mg + everolimus 5 mg:
 - **Combination prolonged PFS over everolimus alone;** lenvatinib alone also beat everolimus.
 - **Combination versus lenvatinib alone: not significantly different.**
 - Basis for FDA approval of **lenvatinib + everolimus after antiangiogenic therapy**; larger confirmation is still needed.

TABLE 144.6 SUMMARY OF SELECTED STUDIES OF TARGETED MULTIKINASE INHIBITORS IN METASTATIC RENAL CELL CARCINOMA

Choueiri *et al.*, 2015, 2017; Motzer *et al.*, 2015, 2018.

Trial	Comparison	Phase	Population	Patients	ORR (%)	Median PFS (months)	Median OS (months)
METEOR	Cabozantinib vs everolimus	III	Progressed on anti-VEGFR therapy	658	21 vs 5 (text); 31 vs 5 (source table)	7.4 vs 3.8	21.4 vs 17.1
CABOSUN	Cabozantinib vs sunitinib	II	Untreated, intermediate/poor risk	157	46 vs 18 (text); 33 vs 12 (source table)	8.2 vs 5.6	Not available
Lenvatinib $\pm$ everolimus	Lenvatinib + everolimus vs everolimus	II	Progressed on anti-VEGFR therapy	101 (these two arms)	43 vs 6	14.6 (text) / 14.1 (table) vs 5.5	—

Where two values are shown, the body text and the source table disagree; the source does not state which analysis each reflects. The lenvatinib phase II enrolled 153 patients across three arms; the table shows two.

- **Pattern:** broadening the kinase spectrum beyond VEGFR (MET, AXL, FGFR) is what finally beat everolimus and sunitinib; **cabozantinib is the only TKI in this chapter with an OS gain over an active comparator (METEOR).**
- **Trap:** CABOSUN is phase II and restricted to intermediate/poor risk.

Tivozanib

- **TKI of VEGFR1–3, c-KIT and PDGFR- β .**
- **TIVO-3** (open-label phase III; relapsed or refractory after 2–3 prior systemic therapies including a VEGFR TKI): tivozanib 1.34 mg daily, 21 of 28 days, versus sorafenib 400 mg twice daily.
 - **PFS 5.6 versus 3.9 months** (HR 0.73); **response 18% versus 8%.**
 - **OS no better** (16.4 versus 19.2 months; HR 0.97).
- **FDA approval March 2021** for advanced RCC after **at least two prior systemic therapies.**

Belzutifan

- **HIF-2 α** , the transcription factor immediately downstream of VHL loss, was long considered **undruggable**; structural work found a targetable binding domain, leading to **HIF-2 α inhibitors.**

- **VHL-associated RCC:** FDA-approved; in the phase II trial almost all patients had tumour shrinkage or stabilisation, with high PFS.
- **LITESPARK-005** (open-label phase III, 746 patients progressing after **both a PD-1/PD-L1 inhibitor and a VEGF TKI**; belzutifan 120 mg versus everolimus 10 mg daily, stratified by IMDC risk and number of prior VEGF TKIs):
 - **PFS improved** (HR 0.75), although **median PFS was identical** (5.6 months each) — **non-proportional hazards**: the curves separate late.
 - OS immature, with **no signal of harm**; patient-reported outcomes favoured belzutifan.
 - **FDA approval for metastatic ccRCC after both PD-1/PD-L1 blockade and VEGF-targeted therapy.**

Inhibitors of the Mammalian Target of Rapamycin

- **mTOR** is an intracellular kinase in several growth-factor signalling cascades that regulates **translation and stability of HIF-1 α** ; growth inhibition by mTOR inhibitors correlates with a block in HIF-1 α translation.
- Two **sirolimus analogues** are active in RCC: **temsirolimus** and **everolimus**.
- **Temsirolimus phase II** (three intravenous weekly doses): modest response; response and PFS **independent of dose**. In the MSKCC poor-risk subgroup, OS was longer than in historical IFN controls → phase III in poor-risk patients.
- **Temsirolimus phase III** (626 previously untreated patients with **three or more predefined poor-risk features, all histological subtypes**): temsirolimus 25 mg IV weekly versus IFN- α (up to 18 million units subcutaneously three times weekly) versus temsirolimus 15 mg + IFN- α 6 million units.
 - **PFS better in both temsirolimus arms** than IFN alone.
 - **OS significantly better with temsirolimus alone** (10.9 versus 7.3 months; $P = 0.008$); **the combination did not improve OS** (8.4 versus 7.3 months).
 - **Toxicity**: mucositis, fatigue, rash, **hyperglycaemia, hypophosphataemia, hypercholesterolaemia**, pulmonary complications — mostly manageable.
 - FDA-approved; a **reasonable front-line choice in poor-risk patients**, though never compared with checkpoint inhibitors.
- **Everolimus** (oral; phase III after sunitinib, sorafenib or both; 10 mg daily versus placebo): stopped early for **superior PFS (4.0 versus 1.9 months)**; **OS not different**. Now supplanted by **nivolumab and cabozantinib** after first-line therapy, but reasonable **in combination with lenvatinib**.
- **Combinations across the VHL pathway** are attractive for efficacy and against resistance, but limited by **overlapping toxicity** requiring major dose reductions.
 - **First-generation VEGFR TKI (sunitinib) + mTOR inhibitor**: too toxic to advance.
 - **Bevacizumab-based combinations** are better tolerated. Bevacizumab + everolimus was active in phase II. **RECORD-2** (bevacizumab + everolimus versus bevacizumab + IFN- α) found similar PFS, OS and safety.
 - **INTORACT** (791 untreated patients; bevacizumab + temsirolimus versus bevacizumab + IFN- α): **no difference in PFS, OS or response**; the temsirolimus combination had better kidney symptom scores (Functional Assessment of Cancer Therapy–Kidney Symptom Index).
 - **Bottom line**: bevacizumab + mTOR inhibitor adds nothing to established first-line standards.

TABLE 144.7 SUMMARY OF SELECTED STUDIES OF MTOR INHIBITORS IN METASTATIC RENAL CELL CARCINOMAHudes *et al.*, 2007; Motzer *et al.*, 2008.

Trial	Comparison	Population	Patients	ORR (%)	Median PFS (months)	Median OS (months)
Temsirolimus phase III	Temsirolimus vs IFN- α vs combination	Poor prognosis, untreated, all histologies	626	8.6 vs 4.8 vs 8.1	5.5 vs 3.1 vs 4.7 (investigator); 3.8 vs 1.9 (independent review)	10.9 vs 7.3 vs 8.4
Everolimus phase III	Everolimus vs placebo	Refractory to VEGF-targeted therapy	410	1 vs 0	4.0 vs 1.9	Not reached vs 8.8

TEMSIROLIMUS VERSUS EVEROLIMUS

	Temsirolimus	Everolimus
Route	Intravenous, weekly	Oral, daily
Setting	First-line, poor risk, all histologies	After VEGF TKI; now mainly with lenvatinib
Registration result	OS gain versus IFN- α	PFS gain versus placebo; no OS gain

- **Trap:** temsirolimus is the only mTOR inhibitor with an OS benefit, and it was proven in **poor-risk** patients, including non-clear cell histology.

Concept — Every targeted agent attacks one consequence of VHL loss: the ligand (bevacizumab), its receptors (TKIs), HIF translation (mTOR) or HIF-2 α itself (belzutifan). Resistance through MET, AXL and FGFR explains why the later TKIs are broader, and none gives the durable remissions of immunotherapy.

VEGF-PATHWAY AGENTS — TARGETS

Agent	Class	Targets
Bevacizumab	Monoclonal antibody	VEGF-A (ligand)
Sorafenib	Oral TKI	VEGFR2, PDGFR- β , raf-1
Sunitinib	Oral TKI	VEGFR2, PDGFR- β , c-KIT, FLT3
Pazopanib	Oral TKI	VEGFR, PDGFR, KIT
Axitinib	Oral TKI	VEGFR1–3 only (highly selective)
Tivozanib	Oral TKI	VEGFR1–3, c-KIT, PDGFR- β
Cabozantinib	Oral TKI	VEGFR, MET, AXL
Lenvatinib	Oral TKI	VEGFR1–3, FGFR1–4 , PDGFR- α , RET, KIT

VEGF-PATHWAY AGENTS — SIGNATURE TOXICITIES

Agent	Toxicities emphasised in the chapter
Bevacizumab	Bleeding, proteinuria , hypertension, fatigue
Sorafenib	Hypertension, fatigue, rash, hand–foot syndrome, diarrhoea
Sunitinib	Diarrhoea, hand–foot syndrome, fatigue, hypertension, marrow suppression (thrombocytopenia), hypothyroidism
Pazopanib	Hepatotoxicity (raised alanine aminotransferase) ; less fatigue and hand–foot syndrome than sunitinib
Axitinib	Diarrhoea, fatigue, hypertension
Cabozantinib	Fatigue, diarrhoea, palmar–plantar erythrodysesthesia ; frequent dose reduction

IMMUNE CHECKPOINT ERA

- The antitumour immune response is set by a balance of **stimulatory and inhibitory signals**. Inhibitory receptors on effector cells, such as CD8+ T lymphocytes, tightly regulate it, and responses are downregulated through **cytotoxic T-lymphocyte-associated antigen-4 (CTLA-4)** and **PD-1**.
- **Ipilimumab** (anti-CTLA-4), single-arm phase II: active in ccRCC, including after IL-2. Autoimmune **enteritis and hypophysitis** were notable, and **autoimmune events predicted response**. Grade 3–4 events in up to a third halted single-agent development.
- **PD-1 and its ligands**: PD-1 on T cells; **PD-L1 and PD-L2 on tumour cells, macrophages and other cells**. Engagement of PD-1 **downregulates antigen-driven immune responses**. Tumours with **PD-1-positive tumour-infiltrating lymphocytes** tend to be larger, higher grade, more advanced and more often **sarcomatoid**.
- **Nivolumab** (anti-PD-1) phase I, with expansion cohorts across solid tumours including RCC: objective responses, several durable, and prolonged stable disease. In the biomarker subset, responses occurred **only in PD-L1-expressing tumours** → PD-L1 proposed as a candidate predictive biomarker. Autoimmune toxicity included diarrhoea, hypophysitis and vitiligo.
- **BMS-936559** (anti-PD-L1) phase I: objective responses and prolonged stable disease in some RCC patients.
- **CheckMate 025** (phase III, 821 patients with a clear cell component after 1–2 antiangiogenic agents; nivolumab 3 mg/kg every 2 weeks versus everolimus 10 mg daily; primary endpoint OS):
 - **OS 25.0 versus 19.6 months** (HR 0.73).
 - **PFS not different** (4.6 versus 4.4 months).
 - **Response 25% versus 5%**.
 - **Grade 3–4 events 19% versus 37%** — nivolumab better tolerated; autoimmune events and fatigue.
- **Bottom line**: single-agent nivolumab is a reasonable option **after VEGF-axis therapy**; this trial prompted front-line testing of PD-1/PD-L1 inhibitors, mostly in combination.

Concept — *Checkpoint inhibitors release a brake on an existing antitumour response rather than stimulating immunity nonspecifically; their toxicity is autoimmune, and their benefit appears as OS and durability rather than PFS.*

IMMUNE CHECKPOINT INHIBITOR-BASED COMBINATION STRATEGIES

- **Rationale**: many patients do not benefit from PD-1/PD-L1 monotherapy, and **complete responses — a surrogate for long-term disease-free survival — occur in fewer than 10%**.
- **CTLA-4** is a **CD28 homologue** on T cells; binding of **B7** costimulatory molecules to CTLA-4 suppresses antigen-dependent T-cell stimulation. Ipilimumab is approved in melanoma, where **PD-1 + CTLA-4 blockade is synergistic but toxic**.
- **CheckMate 016** (phase I dose-finding in RCC): **nivolumab 3 mg/kg + ipilimumab 1 mg/kg every 3 weeks × 4 doses** had the most acceptable toxicity; higher ipilimumab doses gave substantially more grade 3–4 events.
- **CheckMate 214** (phase III, first-line; 1096 randomised to nivolumab + ipilimumab or sunitinib; primary analysis in **847 IMDC intermediate/poor-risk** patients; co-primary OS, PFS and response):
 - **OS: not reached versus 26.0 months** (HR 0.63).
 - **Response 42% versus 27%; complete response 9% versus 1%**.
 - **PFS 11.6 versus 8.4 months** (HR 0.82) — **not significant** against the prespecified threshold.
 - **Toxicity**: grade 3–4 events 46%; immune-related events needing **high-dose glucocorticoids 35%**; permanent discontinuation 22%.

- **Favourable-risk patients did better with sunitinib.**
- Patient-reported outcomes: **fewer symptoms and better health-related quality of life** than sunitinib in intermediate/poor risk.
- → **Standard-of-care option for first-line intermediate/poor-risk ccRCC.**
- **Checkpoint + VEGF-axis combinations — rationale:** single-agent activity of both classes plus preclinical synergy. VEGF-pathway inhibitors are **immunomodulatory**: sunitinib reduces **regulatory T cells**, and sunitinib and sorafenib suppress **immunosuppressive tumour-associated macrophages**.
- **First-generation pairings failed on toxicity:** nivolumab + sunitinib or pazopanib was active in pretreated patients but **prohibitively toxic** → newer, more selective VEGF-pathway agents were paired instead.
- **KEYNOTE-426** (phase III; axitinib 5 mg twice daily + pembrolizumab 200 mg every 3 weeks versus sunitinib; all IMDC risk groups): **PFS and OS both improved** in the intention-to-treat population; **response 60.4% versus 39.6%**. PFS benefit was consistent **across IMDC groups and independent of PD-L1** → **FDA first-line approval, all risk groups, April 2019**.
- **JAVELIN Renal 101** (phase III; avelumab [anti-PD-L1] 10 mg/kg every 2 weeks + axitinib versus sunitinib; primary endpoints PFS and OS in **PD-L1-positive** patients): **PFS improved** in PD-L1-positive patients and in the intention-to-treat population; **OS immature**. Responses were **not differentiated by PD-L1 or tumour mutational burden**. FDA approval May 2019, but **not a preferred regimen in National Comprehensive Cancer Network (NCCN) guidelines** because OS remains under evaluation.
- **CheckMate 9ER** (phase III, untreated, all IMDC groups; cabozantinib 40 mg daily + nivolumab 240 mg every 2 weeks versus sunitinib; primary PFS): **PFS 16.6 versus 8.3 months** (HR 0.51); **response 55.7% versus 27.1%**; OS benefit. Benefits consistent across IMDC groups and **independent of PD-L1**. Final analysis: median **OS 37.7 versus 34.3 months** (HR 0.70) → **FDA approval, all risk groups, January 2021**.
- **CLEAR** (phase III; lenvatinib 20 mg daily + pembrolizumab 200 mg every 3 weeks versus sunitinib, with a third arm of lenvatinib + everolimus): **PFS 23.9 versus 9.2 months** (HR 0.39); **response 71% versus 36.1%**; longer OS (HR 0.66). PFS benefit held **across IMDC groups and regardless of PD-L1, organ metastasis, prior nephrectomy or sarcomatoid histology** → **FDA approval, all risk groups, August 2021**. In the third arm, lenvatinib + everolimus improved PFS over sunitinib (14.7 months; HR 0.65) but **not OS** (HR 1.15).
- **Synthesis:** PD-1/TKI combinations have **never been compared directly with PD-1/CTLA-4 blockade**, but show at least comparable efficacy with a **more favourable toxicity profile** → now the **front-line standard of care**.

TABLE 144.8 PHASE III TRIALS OF IO/IO OR IO/TKI IN ADVANCED CLEAR CELL RENAL CELL CARCINOMAMotzer *et al.*, 2018, 2019, 2021, 2022; Rini *et al.*, 2019. All trials compared against sunitinib.

Trial	Regimen (pairing)	Primary endpoint	Patients	Median PFS, months (HR)	Median OS, months (HR)	ORR (%)
CheckMate 214	Nivolumab + ipilimumab (IO/IO)	OS, PFS, ORR in IMDC intermediate/poor	1096 (847 intermediate/poor)	11.6 vs 8.4 (0.82, NS) → 11.6 vs 8.3 (0.73)	NR vs 26.0 (0.63) → 47.0 vs 26.6 (0.68)	42 vs 27; CR 9 vs 1
KEYNOTE-426	Pembrolizumab + axitinib (IO/TKI)	OS, PFS	861	15.4 vs 11.1 (0.71) → 15.7 vs 11.1 (0.68)	NR vs 35.7 (0.68) → 45.7 vs 40.1 (0.73)	60.4 vs 39.6
JAVELIN Renal 101	Avelumab + axitinib (IO/TKI)	PFS and OS in PD-L1-positive	886 (560 PD-L1-positive)	PD-L1+: 13.8 vs 7.2 (0.61) → 13.8 vs 7.0 (0.62); all patients: 13.8 vs 8.4 (0.69)	PD-L1+: NR vs 28.6 (0.83, NS)	—

Trial	Regimen (pairing)	Primary endpoint	Patients	Median PFS, months (HR)	Median OS, months (HR)	ORR (%)
CheckMate 9ER	Nivolumab + cabozantinib (IO/TKI)	PFS	651	16.6 vs 8.3 (0.51) → 17.0 vs 8.3 (0.52)	NR vs 29.5 → 37.7 vs 34.3 (0.70)	55.7 vs 27.1
CLEAR	Pembrolizumab + lenvatinib (IO/TKI)	PFS, OS	1069 (three arms)	23.9 vs 9.2 (0.39)	NR vs NR (0.66 in text; 0.72 in source table)	71 vs 36.1

Arrows mark successive data cuts (earlier analysis → longer follow-up). IO, immuno-oncology; CR, complete response; NR, not reached; NS, not significant.

- **What the table is for:** the evidence base for every preferred first-line regimen.
- **Pattern:** all five beat sunitinib on PFS or OS. **CLEAR has the largest PFS and response effect.** Nivolumab + ipilimumab is the only IO/IO regimen, the only one confined to intermediate/poor risk, and the only one whose PFS was not significant. JAVELIN is the only one without an OS benefit. **PD-L1 status does not select patients** for any of them.
- **Trap:** avelumab is **anti-PD-L1** (the others anti-PD-1), and avelumab + axitinib is FDA-approved but **not NCCN-preferred**. Nivolumab + ipilimumab is not preferred for favourable risk.

Concept — *The combination era pairs a PD-1 blocker with either a second checkpoint (deeper, more durable responses at the price of autoimmune toxicity) or a VEGF TKI (higher response and PFS, better tolerated); that trade-off, not a difference in survival, drives the choice.*

Other Treatment Options in Patients With Clear Cell Renal Cell Carcinoma

Chemotherapy

- **Conventional cytotoxic chemotherapy is largely ineffective** in ccRCC. Single agents (**5-FU, platinum compounds, gemcitabine, vinblastine, bleomycin**) and a wide range of combinations fail.
- Meta-analyses: **overall response 5.5–6.0%**.
- **Chemoresistance:** overexpression of the **multidrug resistance (MDR) gene** is proposed, but adding MDR inhibitors (**toremifene, ciclosporin, verapamil**) did not help → other, unrecognised factors.
- **No role in most patients with ccRCC.** Possible exception worth study: **sarcomatoid component**, where gemcitabine-based chemotherapy showed promising activity in a small series.

Hormonal Therapy

- Trialled in the **1970s and 1980s**, before cytokines, for lack of alternatives and because the **male preponderance (about 2:1)** suggested a hormonal basis.
- **Medroxyprogesterone** and similar agents produce regressions in a small minority; **response about 2%**.
- **Progestational and other hormonal agents have no current role.**

Concept — *ccRCC is intrinsically chemoresistant and hormone-insensitive; its exploitable vulnerabilities are angiogenic (VHL–HIF) and immunological, which is why effective therapy arrived only with those two drug classes.*

CURRENT THERAPEUTIC STRATEGIES FOR CLEAR CELL RENAL CELL CARCINOMA

First-Line Therapies for Clear Cell Renal Cell Carcinoma

- **Combination immuno-oncology (IO)/TKI or IO/IO therapy** is the preferred first-line approach of both the NCCN and the **European Association of Urology (EAU)**, each **stratifying by IMDC risk** (Table 144.9).

TABLE 144.9 GUIDELINE-BASED FIRST-LINE THERAPIES FOR CLEAR CELL RENAL CELL CARCINOMA — NCCN VERSUS EAU

Risk	NCCN preferred	NCCN other recommended	NCCN useful in certain circumstances	EAU standard of care	EAU if IO not possible
Favourable	Axitinib + pembrolizumab; cabozantinib + nivolumab; lenvatinib + pembrolizumab	Axitinib + avelumab; cabozantinib; ipilimumab + nivolumab; pazopanib; sunitinib	Active surveillance ; axitinib; high-dose IL-2	Axitinib + pembrolizumab; cabozantinib + nivolumab; lenvatinib + pembrolizumab	Sunitinib; pazopanib
Intermediate/poor	The same three + ipilimumab + nivolumab	Cabozantinib; axitinib + avelumab; pazopanib; sunitinib	Axitinib; high-dose IL-2; temsirolimus	The same three + ipilimumab + nivolumab	Cabozantinib ; sunitinib; pazopanib

- **Pattern:** the three IO/TKI combinations are preferred in **every** risk group by both bodies; ipilimumab + nivolumab joins them only for intermediate/poor risk. Single-agent TKIs are fallbacks.
- **Trap:** **active surveillance** is listed only for favourable risk, **temsirolimus** only for intermediate/poor. Single-agent cabozantinib is an NCCN "other recommended" option, not a preferred one.

Subsequent Therapy for Clear Cell Renal Cell Carcinoma

- **NCCN stratifies by prior immunotherapy** and designates **no preferred regimen**; the **EAU stratifies by prior IO versus prior TKI** (Table 144.10).
- **EAU:** after IO → **any VEGF-targeted therapy not previously used** (with the IO); after TKI → **nivolumab or cabozantinib**.

TABLE 144.10 SUBSEQUENT LINE THERAPY FOR CLEAR CELL RENAL CELL CARCINOMA — NCCN VERSUS EAU

Prior therapy	NCCN other recommended	NCCN useful in certain circumstances	EAU standard of care	EAU alternative
IO-naïve (prior TKI)	Axitinib + pembrolizumab; cabozantinib; cabozantinib + nivolumab; ipilimumab + nivolumab; lenvatinib + everolimus; lenvatinib + pembrolizumab; nivolumab	Axitinib; everolimus; pazopanib; sunitinib; tivozanib; belzutifan; bevacizumab; high-dose IL-2; temsirolimus; axitinib + avelumab	Nivolumab; cabozantinib	Axitinib
Prior IO	Axitinib; belzutifan; cabozantinib; lenvatinib + everolimus; tivozanib	Axitinib + pembrolizumab; cabozantinib + nivolumab; everolimus; ipilimumab + nivolumab; lenvatinib + pembrolizumab; pazopanib; sunitinib; bevacizumab; high-dose IL-2; temsirolimus; axitinib + avelumab	Any VEGF-targeted agent not previously used	None

NCCN designates no preferred regimen in either setting. IL-2, interleukin-2; IO, immuno-oncology; TKI, tyrosine kinase inhibitor.

- **Pattern:** after IO, the recommended options are all VEGF-axis agents or belzutifan; after a TKI alone, a checkpoint inhibitor or cabozantinib.
- **Trap:** belzutifan and tivozanib belong to the **post-IO** setting; re-challenge with an IO combination after IO is only "useful in certain circumstances".

Concept — *Sequencing follows one rule: **switch mechanism**. Whichever axis was used first, the next line attacks the other, or a VEGF target not yet used.*

SYSTEMIC THERAPY FOR NON-CLEAR CELL VARIANTS OF RENAL CELL CARCINOMA

- **Non-clear cell subtypes make up about 15–25% of kidney cancers.** They have been studied in few prospective trials and have no agents of proven efficacy, so patients often receive **ccRCC agents**.

Papillary RCC and VEGF-pathway agents

- **VEGFR inhibitors have only modest efficacy in papillary RCC.** Sorafenib and sunitinib showed minimal activity retrospectively; prospective phase II sunitinib responses were **5–10%**.
- **Erlotinib** (oral EGFR inhibitor; Southwest Oncology Group): modest response with poor 6-month PFS.

mTOR inhibitors

- **Temsirolimus:** in an exploratory subgroup of the poor-prognosis phase III trial (about 20% non-clear cell, mostly papillary), OS and PFS were better than with IFN — **limited by being a subgroup analysis**.
- **Everolimus: RAPTOR** (phase II, first-line papillary) gave modest PFS; about a quarter stopped for adverse events (asthenia, fatigue, anaemia). A second phase II in non-clear cell RCC found responders among **chromophobe**, papillary and unclassified tumours, with a **trend to longer PFS in chromophobe → modest activity**.
- **Sunitinib versus everolimus** (two randomised phase II trials): both modest; neither clearly superior; **trend favouring sunitinib**.

Hereditary papillary renal carcinoma and the MET pathway

- Hereditary forms of papillary and chromophobe RCC reveal the **critical molecular events in both familial and sporadic tumours**, as VHL did for ccRCC.
- **Hereditary papillary renal carcinoma (HPRC):** bilateral, multifocal **papillary type 1 RCC**, caused by **activating germline mutations in the tyrosine kinase domain of the proto-oncogene MET on chromosome 7**, usually with **nonrandom duplication of the chromosome carrying the mutant allele**.
- **MET** is the cell-surface receptor for **hepatocyte growth factor (HGF)**; kinase-domain mutations make it **constitutively active**. The HGF/MET pathway drives growth, proliferation and motility.
- **Fig. 144.12:** MET activation (by HGF binding, autocrine or paracrine, or by activating mutation or overexpression) signals through effectors such as **Grb2** → motility, matrix remodelling, survival and proliferation → invasion, metastasis and tumour progression. Intervention points: **HGF-binding antagonists, MET tyrosine kinase inhibitors, receptor/effector antagonists**.
- **Sporadic papillary RCC:** activating somatic **MET** mutations in a **minority (about 13%)**; **trisomy 7** (carrying both *HGF* and *MET*) in **more than two-thirds** — an alternative route to MET activation.
- **Foretinib** (MET + VEGFR2 inhibitor; intermittent and continuous dosing cohorts): modest overall response; **primary endpoint (response ≥25%) not met**; hypertension commonest adverse event. **Germline MET mutation carriers: about half responded**, versus few without → MET inhibition viable in a subset.
- **Selective MET inhibitors** (savolitinib, crizotinib; little VEGFR activity) show activity in MET-activated tumours; **randomised trials against sunitinib are ongoing**.

Hereditary leiomyomatosis and renal cell cancer and fumarate hydratase

- **HLRCC:** germline mutation of **fumarate hydratase (FH)**, the Krebs-cycle enzyme converting fumarate to malate → **highly aggressive papillary type 2 RCC**.
- **Fumarate accumulation** causes **VHL-independent HIF upregulation** and activation of proangiogenic and growth factors.
- **Metabolic shift:** Krebs-cycle disruption → dependence on **aerobic glycolysis** → exquisite sensitivity to **glucose deprivation**.

- **Bevacizumab + erlotinib** (NCI) was designed to restrict glucose delivery to the tumour microenvironment. Interim data: high response rates, with partial responses in **about two-thirds of HLRCC patients versus under one-third with sporadic papillary RCC**; longer follow-up required.

Collecting duct carcinoma

- Rare variant resembling **urothelial carcinoma**; **gemcitabine + carboplatin** gives modest response and survival.
- **Summary: no standard approach of proven efficacy** for most non-clear cell RCC; **trial enrolment should be considered for all**.

HEREDITARY RCC SYNDROMES — GENE, TUMOUR AND THERAPEUTIC LEAD

Syndrome	Gene (chromosome)	Tumour	Mechanism	Therapeutic lead
Von Hippel-Lindau	<i>VHL</i> (3p), loss of function	Bilateral multifocal clear cell	HIF- α not degraded	VEGF/VEGFR, mTOR, HIF-2α (belzutifan)
Hereditary papillary renal carcinoma	<i>MET</i> (7), activating	Bilateral multifocal papillary type 1	Constitutive MET kinase	MET inhibitors (foretinib, savolitinib, crizotinib)
Hereditary leiomyomatosis and RCC	<i>FH</i> , loss of function	Aggressive papillary type 2	Fumarate $\rightarrow$ HIF; aerobic glycolysis	Bevacizumab + erlotinib (glucose restriction)

- **Hereditary versus sporadic papillary:** germline *MET* mutation in HPRC versus somatic mutation in a minority, with trisomy 7 in the majority, of sporadic tumours; responses to MET inhibition and to bevacizumab + erlotinib are markedly greater in the hereditary form.

Concept — *In non-clear cell RCC the hereditary syndrome names the driver, and the driver names the drug: MET inhibition for MET-driven papillary tumours, glucose restriction for FH-deficient tumours. Borrowed clear cell drugs give only modest activity.*